

Effect of Bathymetric Curvature on Gulf Stream Instability in the Vicinity of the Charleston Bump

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ABSTRACT

The effect of the isobathic curvature on the development and evolution of Gulf Stream frontal waves (meanders and eddies) in the vicinity of the Charleston Bump (a topographic rise on the upper slope off Charleston, South Carolina; referred to as CB hereinafter) is studied using the Hybrid-Coordinate Ocean Model (HYCOM). Baroclinic and barotropic energy transfers from the Gulf Stream to its meanders and eddies that appear as cold and warm anomalies are computed for four different cases. In case I, the curvature of the isobaths is artificially reduced and the CB is removed from the bathymetry. In this simulation, the simulated Gulf Stream meanders were barely noticeable in the study region. Energy transfer from the Gulf Stream to meanders and eddies was negligible. In case II, the curvature of the isobaths was the same as in case I, but a bump of the scale of the CB was added to the bathymetry. In this simulation, Gulf Stream meanders were amplified while passing over the CB. In case III, the CB was removed from the bathymetry as in case I, but the curvature of the isobaths was similar to the actual bathymetry, which was larger than that of cases I and II. In this simulation, large meanders were simulated, but the development of these meanders was not confined to the region of the CB. The total baroclinic and barotropic energy transfer rate in this case was an order of magnitude greater than in case II, suggesting that isobathic curvature was able to generate Gulf Stream meanders and eddies even without the presence of the CB. In case IV, actual bathymetry data, which contain both the CB and the isobathic curvature, were used. In this case, large-amplitude Gulf Stream meanders were simulated and there was also a tendency for the amplification of the meanders to be anchored downstream of the CB, consistent with observations. The results from this study suggest that the formation of the “Charleston Trough,” a Gulf Stream meander that appears as a low pressure or depressed water surface region downstream of the bump, is the result of the combined effect of the CB and the isobathic curvature in the region. The isobathic curvature plays a major role in enhancing the baroclinic and barotropic energy transfer rates, whereas the bump provided a localized mechanism to maximize the energy transfer rate downstream of the CB.

1. Introduction

It was revealed by early sea surface temperature (SST) satellite radiometer data and derived images that wavelike perturbations traveled northward along the western edge of the Gulf Stream front at a speed of about 30 km day^{-1} (Legeckis 1975, 1979) in the South Atlantic Bight (SAB; Fig. 1). Northward propagation of similar wavelike patterns was also evident in Lee and Mayer's (1977) current measurements in the Straits of

Florida; off Onslow Bay, North Carolina (Pietrafesa and Janowitz 1980; Bane et al. 1981); and in the Georgia Bight Experiment I (GABEX I) moored array (Lee and Atkinson 1983). These wavelike perturbations were believed to be manifestations of Gulf Stream lateral meanders and eddies. In fact, Pietrafesa and Janowitz (1980) provided phase speed, $25\text{--}75 \text{ cm s}^{-1}$, and wavelength, $100\text{--}300 \text{ km}$, estimates of the frontal waves as they passed by mooring current meters in Onslow Bay.

At approximately 32°N , a submarine ridge called the Charleston Bump (referred to as CB hereinafter) extends seaward from the continental slope offshore of Charleston, South Carolina (Fig. 1). This topographic

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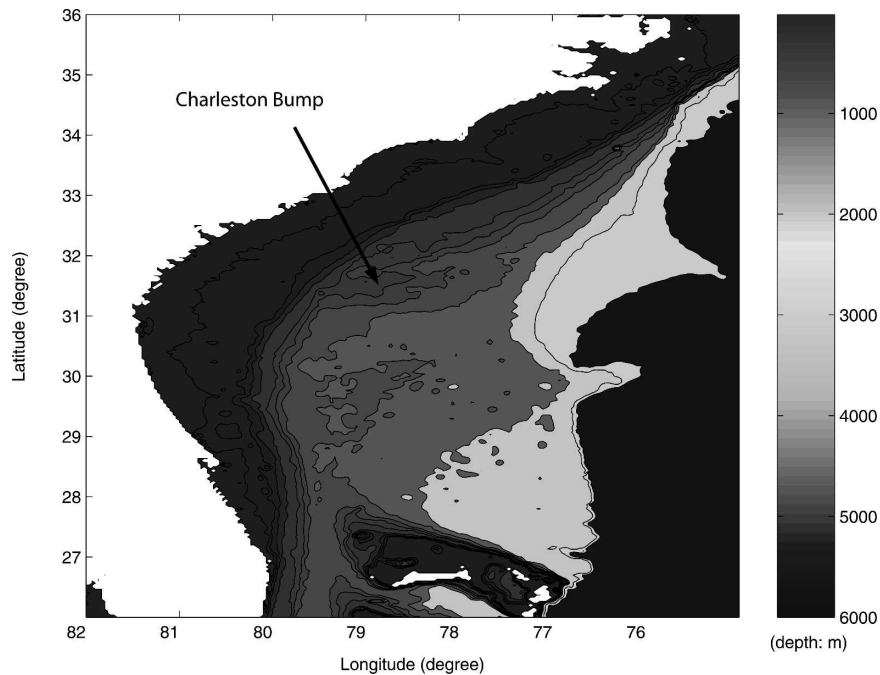


FIG. 1. Bathymetry of South Atlantic Bight from ETOPO2.

feature produces an eastward deflection in the Gulf Stream (Pietrafesa et al. 1978; Rooney et al. 1978; Brooks and Bane 1978), forming a quasi-permanent 10–50-km meander. It has also been reported that the lateral meanderings of the Gulf Stream undergo a dramatic “amplification” (Knauss 1969; Maul et al. 1978; Bane and Brooks 1979; Bane 1983; Hood and Bane 1983; Olson et al. 1983) downstream from the CB. Accumulating evidence has suggested a regional difference in spatial scale of Gulf Stream meanders: south of the latitude of the CB cold-core frontal eddies tend to occur as smaller, elongated filaments (Lee and Mayer 1977; Lee et al. 1981; Lee and Atkinson 1983) and north of CB frontal eddies tend to be considerably larger (Pietrafesa and Janowitz 1979, 1980; Bane and Brooks 1979; Pietrafesa and Janowitz 1980; Bane et al. 1981; Brooks and Bane 1981, 1983). A recent example consistent with these prior observations is presented in Fig. 2, which shows the Moderate Resolution Imaging Spectroradiometer (MODIS) SST distribution throughout the SAB from 12 to 17 January 2004. A perturbation on 12 January 2004 at 32°N was amplified after passing over the CB.

The triggering mechanism for the larger meanders and filaments downstream of the CB is presumably the topographic effect of the CB (Pietrafesa et al. 1978; Rooney et al. 1978; Brooks and Bane 1978; Chao and Janowitz 1979; Legeckis 1979; Pietrafesa 1983; Dewar and Bane 1985). Since Gulf Stream flow has both hori-

zontal and vertical shear, many studies surmise that the large meanders result from local barotropic and baroclinic instabilities. In a two-layer model of the Gulf Stream, Orlanski (1969) demonstrated that topographical slope is a stabilizing factor, while the height of the topography is a destabilizing factor. Sun and Pietrafesa (1992) extended the Orlanski model by including both laterally and vertically sheared horizontal currents in a two-layer Gulf Stream model and showed that energy is transferred between the eddy and the mean flow in both barotropic and baroclinic processes. Xue and Mellor (1993) solved the eigenvalue problem for an instability study of the Gulf Stream, using two analytical cross sections representing the mean conditions of the Gulf Stream upstream and downstream of the CB, and found that the most unstable solution in the region downstream of the CB has a slightly longer wavelength and slower phase speed than those in the region upstream of the bump. Dewar and Bane (1985) computed the energy budgets of the eddies and the mean flow in the Gulf Stream near the CB using observations from Deflection and Meander Energetics Experiment (DAMEX) current-meter mooring deployments, hydrographic surveys, and AXBT surveys and found that immediately north of the CB the flow appears to be both barotropically and baroclinically unstable. Miller and Lee (1995a) provided a thorough review on the stability of Gulf Stream meanders in the SAB. They conducted a series of numerical simulations of the Gulf

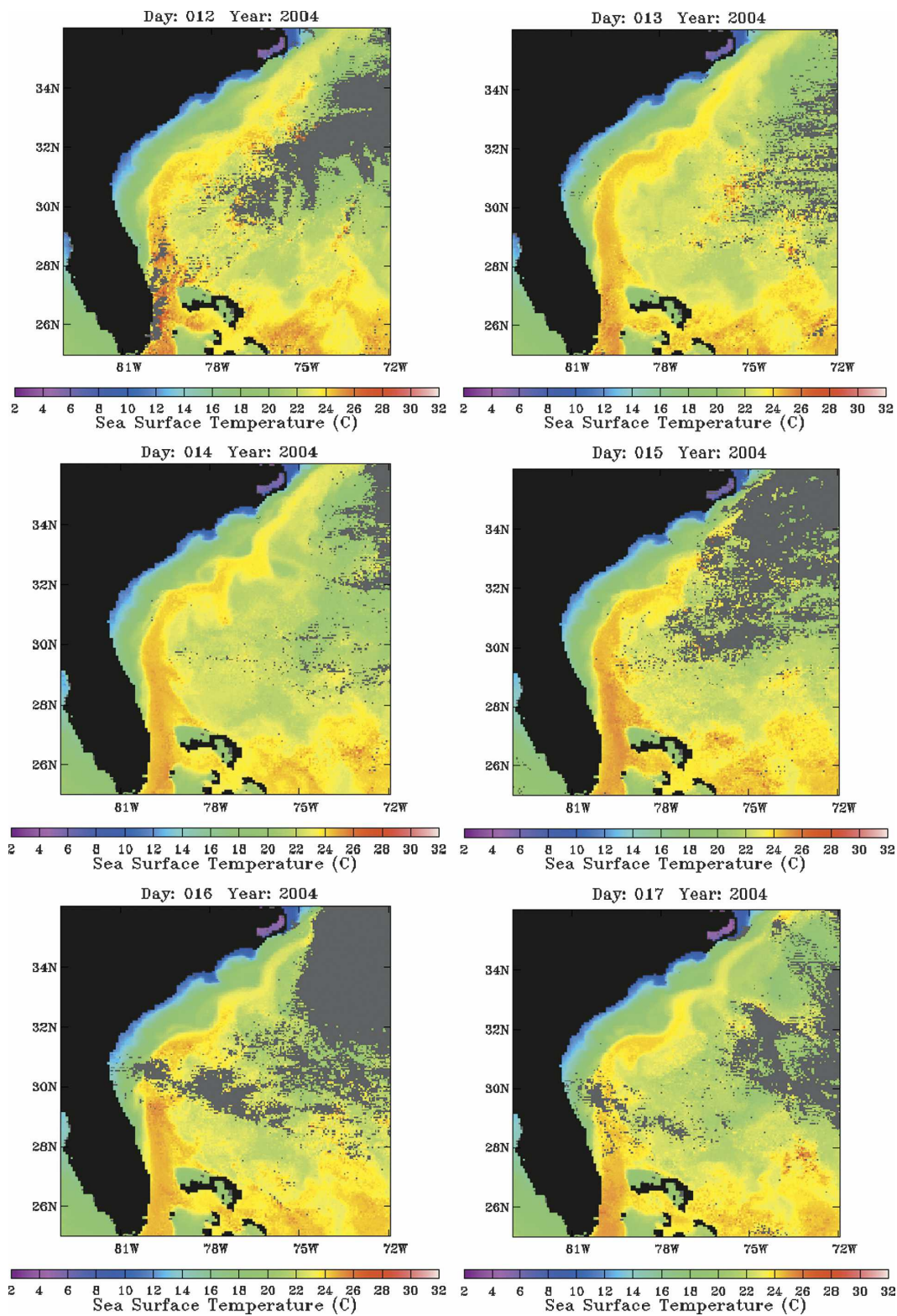


FIG. 2. MODIS SST of SAB from 12 to 17 Jan 2004. An eddy on the west edge of the Gulf Stream at 32°N is amplified.

Stream meanders in the SAB using the Princeton Ocean Model and analyzed the energetics of the simulated circulation. They showed that a hybrid baroclinic–barotropic instability of the Gulf Stream flow was responsible for the growth of meanders. Miller and Lee (1995b) further examined the dynamics of a particular Gulf Stream meander event produced in the SAB; they conclude that geostrophic balance dominated the along-isobath flow and the nonlinear advection terms in the horizontal momentum equation played an important role in the spatial–temporal development of the Gulf Stream meander.

Another characteristic of the bathymetry in the SAB is the curvature of the isobaths (Fig. 1). The isobaths are north–south oriented south of 30°N and turn sharply to northeast–southwest oriented around 31°N. The CB is located in the region of large curve where the Gulf Stream turns from northward to northeastward. So, the question is, could the curvature of the isobaths have an effect on development of Gulf Stream meanders and eddies? The radius of curvature can be as small as 200–300 km, and the centrifugal force required for a current of approximately 2 m s^{-1} is of magnitude 10^{-5} N , which is the same order of magnitude of the local Coriolis force. The vorticity created by the curvature is of the order of 10^{-5} s^{-1} , which is of same order of magnitude of the planetary vorticity. Thus, two hypotheses to be tested are 1) does isobathic curvature have a significant effect on the stability of the Gulf Stream and 2) are the combined effects of isobaths curvature and the influence of the CB responsible for the development of meanders in the vicinity of the CB.

The significance of understanding the amplification of Gulf Stream meanders and eddies in the vicinity of the CB is threefold. First, large meanders and eddies propagating downstream of the CB have great influence on coastal circulation and cross-shelf transport. Lee et al. (1989) studied the response of South Carolina shelf waters to wind and Gulf Stream forcing with the aid of a comprehensive set of atmospheric and oceanic observations collected during the Genesis of Atlantic Lows Experiment (GALE). They showed that the Gulf Stream appears to have two preferred modes, either onshore or offshore, and the subtidal variability in the outer shelf is dominated by synoptic frontal eddies and meanders when the stream is in an onshore mode. Savidge et al. (1992) found that the offshore position of the Gulf Stream front appears to have facilitated the influence of the Gulf Stream on the shelf waters all across the continental shelf. Second, Gulf Stream frontal waves have a significant influence on productivity of the southeast U.S. continental shelf, as demonstrated by Atkinson et al. (1982) and Hoffmann et al. (1980).

These pioneering works demonstrated that without the onshore flux of nutrient laden Gulf Stream frontal wave waters, the North Carolina continental shelf would be a biological desert. Lee et al. (1991) stated that the onshore transport of new nitrogen from the nutrient-bearing strata beneath the Gulf Stream indicates that frontal eddies serve as a “nutrient pump” for the shelf. New nitrogen flux to the shelf due to Gulf Stream input could support production of $7.4 \times 10^{12} \text{ gC yr}^{-1}$, and the continental shelf area upstream from the CB was identified as a high carbon production area, while there is net nutrient loss downstream from CB. Finally, the warm filament associated with the large eddies downstream of the bump can transport *Karenia brevis* cells from the Gulf Stream to the inshore area. Pietrafesa et al. (1988), Tester et al. (1991), and Tester and Steidinger (1997) found that the red tide bloom that occurred in the fall of 1987 in Onslow Bay was caused by the across-shelf transport of red tide laden Gulf Stream frontal filament waters driven across the continental shelf by persistent northeasterly winds. Clearly these frontal waves and their interaction with bottom topography and atmospheric winds are biologically important throughout the SAB. To better understand the dynamics of these frontal waves in the vicinity of the CB, we now attempt to numerically simulate the processes associated with their formation and amplification.

In the present study, a series of numerical experiments were carried out to study the effect of bathymetry on the development and evolution of the Gulf Stream meanders and eddies downstream from the CB in terms of both the bottom bump feature and the curvature of the isobaths. The numerical model used in this study and the set of experiments conducted are described in section 2. Results are presented in section 3. Section 4 presents the diagnostic results of the energetics of the simulated Gulf Stream, followed by a conclusion section.

2. Brief description of the numerical model and numerical experiments

a. Description of the numerical model and data

The Hybrid-Coordinate Ocean Model (HYCOM) (Bleck 2002) is used in this study. HYCOM is a primitive equation ocean general circulation model that evolved from the Miami Isopycnic-Coordinate Ocean Model (MICOM) (Bleck et al. 1992). HYCOM was developed to address known shortcomings of the MICOM vertical coordinate scheme. Vertical coordinates in HYCOM remain isopycnic in the open, stratified ocean. However, they smoothly transition to z coordinates in the weakly stratified upper-ocean mixed

layer, to terrain-following sigma coordinates in shallow water regions, and back to level coordinates in very shallow water. The latter transition prevents layers from becoming too thin where the water is very shallow. HYCOM has been widely used to study the North Atlantic circulation and Gulf Stream variability (Chassignet et al. 2003, and references therein) and the Gulf of Mexico Loop Current and its eddy-shedding processes (Romanou et al. 2004).

In this study, the HYCOM model will be driven by the following atmospheric forcing fields at the ocean surface: wind, air temperature, relative humidity, short-wave radiation, and longwave radiation. To resolve small-scale perturbations of the Gulf Stream front, a nested grid covering the North Atlantic and then scaled down to the SAB will be implemented. The outer domain covers the North Atlantic (17°S – 52°N , 98°W – 14°E) while the inner domain covers the SAB (26° – 36°N , 82° – 71°W). The horizontal resolution for the outer and inner domains is 0.18° and 0.06° , respectively, and the vertical resolution is 22 layers that stretch or shrink vertically as a function of total depth. The bathymetry is derived from the ETOP2 database. The outer domain North Atlantic model will be spun up for 1 year with an initial condition constructed from the Levitus dataset (Levitus 1982) and forced by Comprehensive Ocean Atmosphere Data Set (COADS) monthly climatology atmospheric forcing, and the result will be used as the initial and boundary conditions of the inner domain simulations.

b. Numerical experiments

The purpose of the study is to investigate the topographic effect of the CB and the curvature of the isobaths in the vicinity of the CB on the amplification of smaller perturbation on the shoreward edge of the Gulf Stream front. To study the effect of the curvature of the isobaths two sets of bathymetries, one with isobathic curvature that is less than what actually occurs (Figs. 3a, b) and the other with isobathic curvature similar to the realistic bathymetry (Figs. 3c,d), will be used in the simulations and the results will be compared. In each of the two sets of bathymetries, two, one without a bump (Figs. 3a,c) and the other with a bump (Figs. 3b,d), will be used in the simulations and the results will be compared to assess the effect of the bump for different isobathic curvature settings. A summary of the simulations is given in Table 1.

3. Results

In this section, the results from the four inner-domain simulations will be presented and compared. SST

and temperature anomaly at 125-m depth will be shown in sections 3a and 3b, respectively.

a. Manifestation of Gulf Stream meanders depicted by SST

The evolution of the SST of case I from 17 January to 3 February is shown in Figs. 4Ia–Ic. In this simulation, isobathic curvature is smaller than that of the actual isobaths, and the horizontal isotherms approximately follow the isobaths. Disturbances on the shoreward edge, or west wall, of the Gulf Stream are barely noticeable in this simulation, suggesting that the Gulf Stream is stable.

The SST of case II from 17 January to 3 February is shown in Figs. 4IIa–IIc. In this simulation, the curvature is the same as in case I, but a bump is added to the bathymetry. A significant Gulf Stream meander occurred downstream of the bump near 32°N from 21 January to 25 January, and a warm filament appeared on the shoreward edge of a Gulf Stream eddy. It intensified after passing over the bump.

The SST of case III from 17 January to 3 February is shown in Figs. 4IIIa–IIIC. The bathymetry in this case contains no CB as in case I, but the isobathic curvature is larger than that in case I. Gulf Stream frontal perturbations traveling northeastward on the west edge of the Gulf Stream front can be clearly seen in these figures. Since there is no bump in the bathymetry and the Gulf Stream is initialized the same way as in case I, the difference between Figs. 4Ia–Ic and IIIa–IIIC indicates the effect of the curvature of the isobaths on the instability of the Gulf Stream. Clearly, the increase in isobathic curvature in the region near 30° – 32°N in case III resulted in well-defined meanders that were not present in case I.

The SST of case IV from 17 January to 3 February is shown in Figs. 4IVa–IVc. In this simulation, a large meander/eddy pair occurred downstream of the CB near 32°N from 25 January to 3 February, and the scale of the meander/eddy is much larger than the one in cases II and III. Both the scale and the shape are comparable to the satellite observations shown in Fig. 2. The frontal feature is also traveling northeastward, but at a slower speed than that in cases II and III. Comparing the results among cases I–IV, it becomes evident that realistic Gulf Stream meanders are reproduced only when both realistic isobathic curvature and the CB are considered.

b. Temperature anomaly at 125-m depth

The propagation and amplification of the Gulf Stream meanders and eddies can be seen more clearly

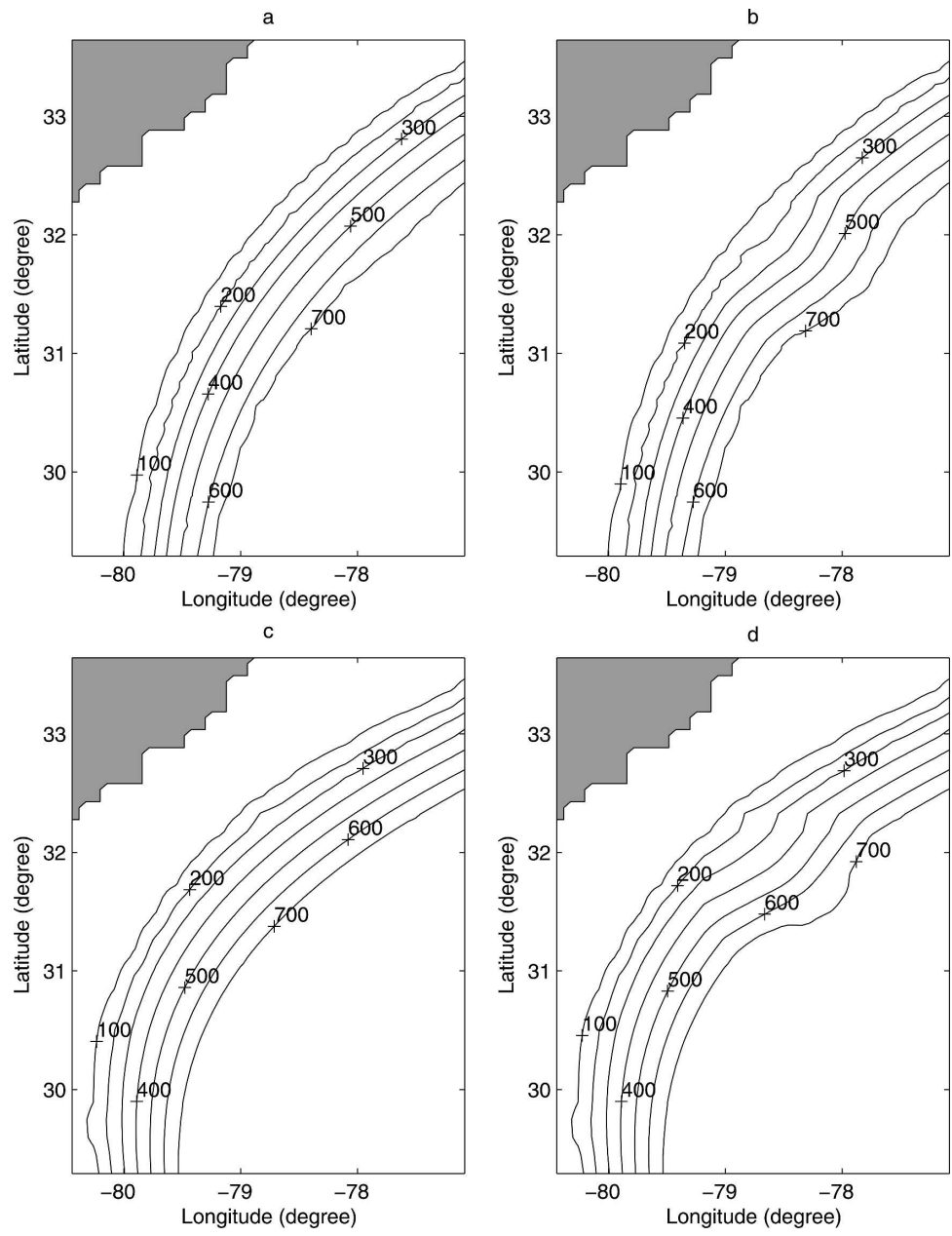


FIG. 3. Bathymetry contour: (a) small curvature without a bump, (b) small curvature with a bump, (c) large curvature without a bump, and (d) large curvature with a bump (units: m).

TABLE 1. List of numerical simulations.

Simulation	Curvature	Bump	Bathymetry	Description
I	Small	No	Fig. 3a	Small curvature without bump
II	Small	Yes	Fig. 3b	Effect of bump alone
III	Large	No	Fig. 3c	Effect of curvature alone
IV	Large	Yes	Fig. 3d	Combined effect of bump and curvature

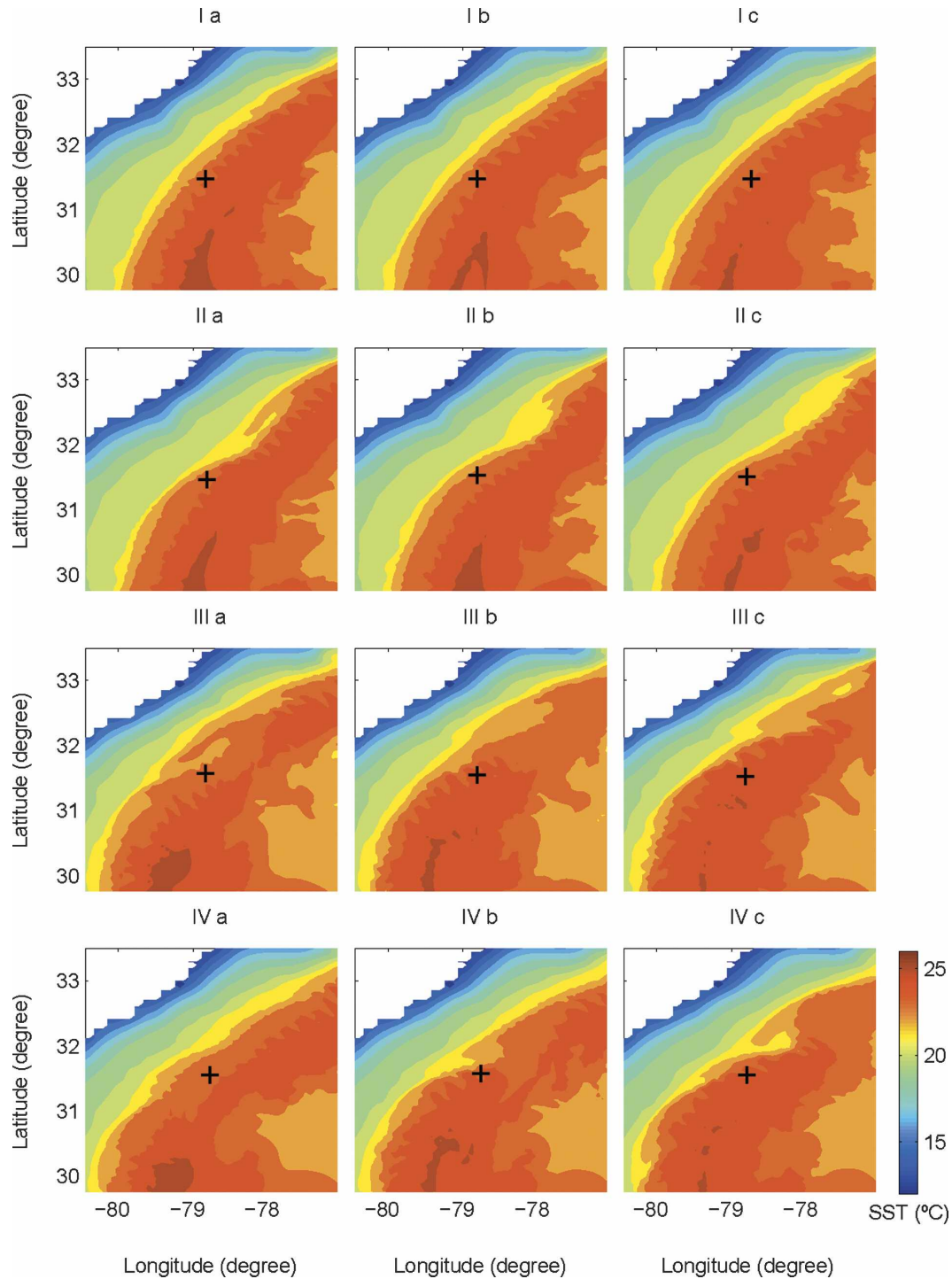


FIG. 4. SST (7.5-m depth) of 23, 25, and 27 Jan for (Ia)–(Ic) case I, (IIa)–(IIc) case II, (IIIa)–(IIIc) case III, and (IVa)–(IVc) case IV.

in the temperature anomaly field below the mixed layer than at the sea surface. The temperature anomaly field at 125-m depth from 17 January to 3 February of cases I, II, III, and IV is shown in Figs. 5Ia–IVc. In case I,

there were only very weak cold and warm anomalies ($-1^{\circ}\text{C} < T < 1^{\circ}\text{C}$) on the shoreward edge of the Gulf Stream (Figs. 5Ia–Ic). They travel northeastward along the isobaths, but are not amplified during this propa-

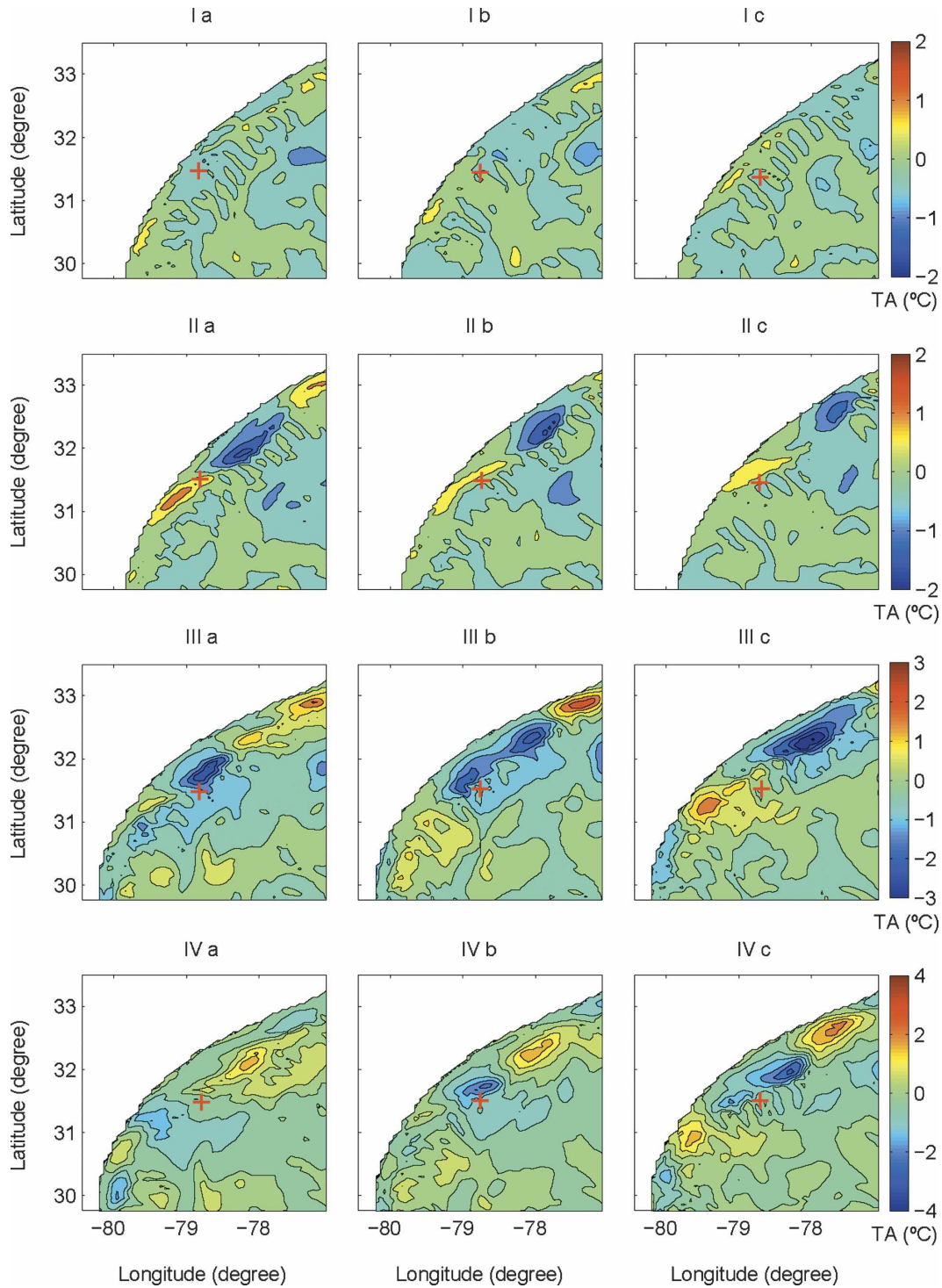


FIG. 5. Water temperature anomaly (TA) at 125-m depth of 23, 25, and 27 Jan for (Ia)–(Ic) case I, (IIa)–(IIc) case II, (IIIa)–(IIIc) case III, and (IVa)–(IVc) case IV.

gation. In case II, a cold anomaly originally located at 32.2°N on 17 January was amplified after passing over the CB. Because of the bump, the cold anomaly was deflected seaward. The warm anomaly that followed

this cold anomaly also increased, the range of the cold and warm anomaly was -2° to 2°C , about 2 times as large as that in case I. The difference between case I and case II depicts the effect of the CB.

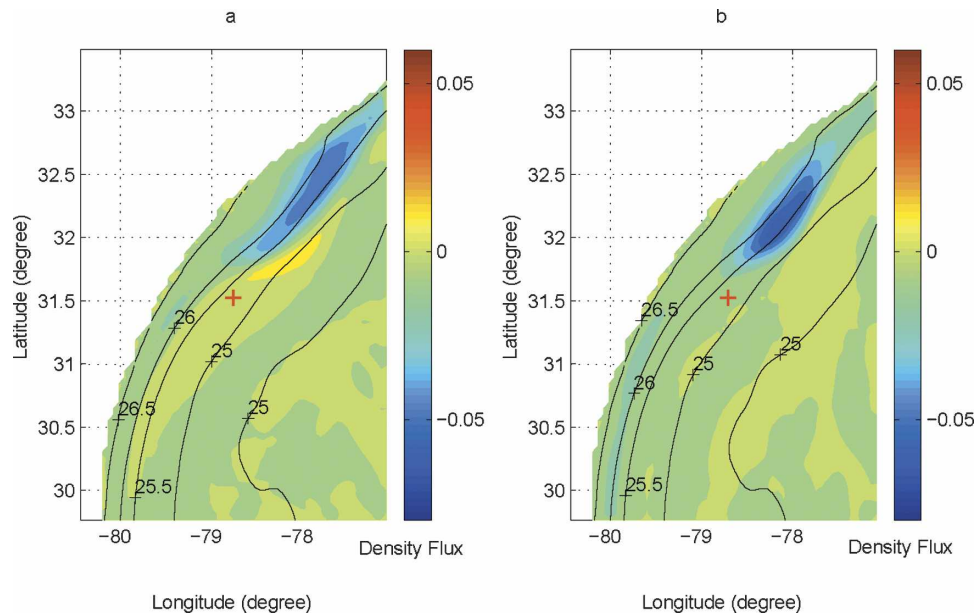


FIG. 6. (a) The x component of density flux $\overline{\rho' u'}$; (b) y component of density flux $\overline{\rho' v'}$ of case IV at 125-m depth (units: $\text{kg m}^{-2} \text{s}^{-1}$).

In case III, there were cold and warm anomalies propagating northeastward along the Gulf Stream front at 125-m depth and they were amplified even though there is no bump in the bathymetry for this simulation. The range of the cold/warm anomaly was -3° to 3°C , which was larger than that in both cases I and II. This suggests that isobathic curvature played a significant role in the development of Gulf Stream meanders.

In case IV, a cold anomaly developed near 31.2°N on 23 January and was significantly amplified after passing over the CB. The minimum of the cold anomaly reached -4°C . The difference between case IV and case III was the bump effect. The difference between case IV and case I was the combined effect of the bump and the isobathic curvature. It is evident that the combined effect of isobathic curvature and the CB produced the largest temperature anomalies below the mixed layer among all four cases. The Gulf Stream meanders depicted at 125-m depth is consistent with the results depicted by SST.

The question then is: How does isobathic topography affect either the attenuation or amplification of Gulf Stream meanders and eddies? The assumption is that the topography affects baroclinic and barotropic process. According to Pedlosky (1979), perturbation flow can extract energy from the mean flow via the $\overline{u'v'}$ (barotropic) and $\overline{u'\rho'}$, $\overline{v'\rho'}$ (baroclinic) processes. Further, Janowitz and Pietrafesa (1982) showed that bottom topography can kinematically induce onshore (off-shore) motions when the isobaths spread; that is, the

cross-shelf slope decreases or the topography deepens (converge, i.e., the cross-shelf slope increases or the topography shallows) in the direction of flow of both barotropic and baroclinic boundary currents. Figures 6a and 6b show the density fluxes ($\overline{u'\rho'}$, $\overline{v'\rho'}$) at 125-m depth. It can be seen clearly that in a large area around the cold anomaly, the ($\overline{u'\rho'}$, $\overline{v'\rho'}$) terms were positive, so that there was release of potential energy from the mean flow to the eddy, which indicates the presence of a baroclinic instability. Figures 7a–c show the momentum flux ($\overline{u'v'}$) superimposed on the mean flow. The positive value of $\overline{u'v'}$ indicates that energy was transferred from the mean flow to the perturbation through barotropic processes.

Since the cold anomaly is associated with cyclonic vorticity and the warm anomaly is associated with anticyclonic vorticity, both the topography and the curvature of the isobaths tend to affect the cold and warm anomalies due to the conservation of potential vorticity. From SST and the temperature anomaly at 125-m depth, we can see that both the CB and the curvature tend to amplify the cold/warm anomaly. The combination of the CB and the curvature produced the largest impact on the amplification of the cold/warm anomalies, as shown in case IV. Thus, it seems that both isobathic curvature and the CB played a significant role in inducing the baroclinic and barotropic stability of the Gulf Stream. In the next section, more in-depth energy analysis will be carried out to identify barotropic and baroclinic processes.

4. Diagnostics of energy conversion

In this section, an energy analysis will be applied in case IV using two methods. The first method is based on traditional eddy kinetic and potential energy equations. This method was used for analyzing the energetics of the Florida Current (Brooks and Niiler 1977) and Gulf Stream after it has passed Cape Hatteras (Rossby 1987). The second method is based on a Multiscale Energy and Vorticity Analysis (MS-EVA) (Liang and Robinson 2005a, 2005b, manuscript submitted to *Dyn. Atmos. Oceans*, hereinafter LR05). The result of the first method is discussed in section 4a and the second method in section 4b. MS-EVA will also be applied to cases II and III, and the results will be discussed in section 4c.

a. Energy analysis using Brooks and Niiler (1977) energy equation for case IV

Consider, first, the energy balance in the simulated circulation using the energy balance equation of Brooks and Niiler (1977) and Rossby (1987), as shown in Eq. (1) [here (1a)–(1f) refer to terms within Eq. (1) rather than to separate equations]:

$$\frac{d}{dt} \left\{ \frac{1}{2} (\overline{u'^2} + \overline{v'^2}) + \frac{1}{2} g \overline{\rho'^2} \right\} / \left| \frac{\partial \bar{\rho}}{\partial z} \right| \rho_0 = \quad (1a)$$

$$- \left\{ \frac{\partial}{\partial x} (\overline{u'p'}) / \rho_0 + \frac{\partial}{\partial y} (\overline{v'p'}) / \rho_0 + \frac{\partial}{\partial z} (\overline{w'p'}) / \rho_0 \right\} \quad (1b)$$

$$- \left\{ \overline{u'^2} \frac{\partial \bar{u}}{\partial x} + \overline{v'^2} \frac{\partial \bar{v}}{\partial y} + \overline{u'v'} \left(\frac{\partial \bar{v}}{\partial x} + \frac{\partial \bar{u}}{\partial y} \right) \right\} \quad (1c)$$

$$- \left\{ g \overline{u'\rho'} \frac{\partial \bar{\rho}}{\partial x} / \left| \frac{\partial \bar{\rho}}{\partial z} \right| \rho_0 + g \overline{v'\rho'} \frac{\partial \bar{\rho}}{\partial y} / \left| \frac{\partial \bar{\rho}}{\partial z} \right| \rho_0 \right\} \quad (1d)$$

$$- \left\{ \overline{u'w'} \frac{\partial \bar{u}}{\partial z} + \overline{v'w'} \frac{\partial \bar{v}}{\partial z} \right\} \quad (1e)$$

$$- \left\{ \frac{1}{2} u' \left(u'^2 + v'^2 + g \rho'^2 \right) / \left| \frac{\partial \bar{\rho}}{\partial z} \right| \rho_0 \right\}_x + \frac{1}{2} v' \left(u'^2 + v'^2 + g \rho'^2 \right) / \left| \frac{\partial \bar{\rho}}{\partial z} \right| \rho_0 \right\}_y. \quad (1f)$$

The first term on the right-hand side [(1b)] represents the pressure-work term; the second term [(1c)] represents the conversion of perturbation kinetic energy; the third term [(1d)] represents the conversion of perturbation potential energy from the mean kinetic and potential energy, respectively. The fourth term [(1e)] represents the energy transfer due to small-scale shear instabilities (Brooks and Niiler 1977), and the fifth term [(1f)] represents the conversion of perturbation kinetic and potential energy by the perturbation

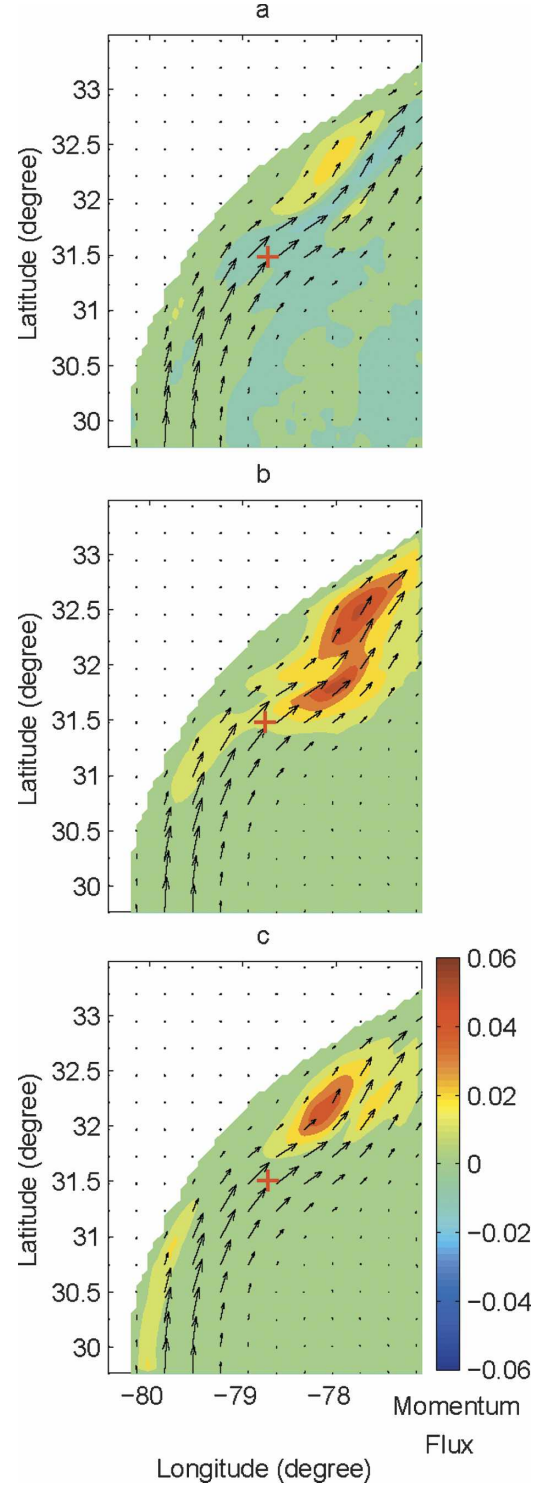


FIG. 7. Momentum fluxes: (a) $\overline{u'v'}$, (b) $\overline{u'u'}$, and (c) $\overline{v'v'}$ of case IV at 125-m depth (units: $\text{m}^2 \text{s}^{-2}$).

field. Dewar and Bane (1985) analyzed the eddy energy in the CB region based on observations from the DAMEX project. In the analysis, energy balance terms were nearly identical to terms (1b), (1c), and (1d), with only two slight differences. First, Eq. (1) includes both eddy kinetic energy and eddy potential energy so that the total eddy energy decay and growth can be estimated, whereas Dewar and Bane (1985) considered eddy kinetic and potential energies separately. Second, terms (1b), (1c), and (1d) are equal to the corresponding terms in Dewar and Bane (1985) divided by water density. Note that Dewar and Bane (1985) estimated each term in centimeters–gram–second (cgs) units (i.e., $\text{ergs cm}^{-3} \text{ s}^{-1}$), whereas meters–kilogram–second (MKS) units are used in our calculations. Therefore, the energy unit used in our calculation differs from Dewar and Bane's by a factor of 10^4 (assuming water density is approximately 1000 kg m^{-3}). The values of the four terms (1a), (1b), (1c), and (1d), which are the major terms of Eq. (1), are presented below.

Figure 8 shows the baroclinic transfer rate term [term (1d)] of case IV at depth of 7.5, 25, 47.5, 80, 125, 190, 305, 495, and 760 m. In this figure, it can be seen that the largest transfer rate is located at 80-m and 125-m depth. The location of the largest transfer rate is at approximately (32°N , 78.5°W), immediately north of the CB. The maximum value of the baroclinic transfer rate is $5 \times 10^{-7} \text{ m}^2 \text{ s}^{-3}$, consistent with the calculations of Dewar and Bane (1985). The baroclinic transfer rate farther downstream of the CB around (32.7°N , 77.5°W) is negative, indicating that potential energy is transferred back to the mean flow in that region.

Figure 9 shows the barotropic transfer rate [term (1c)] of case IV at depth of 7.5, 25, 47.5, 80, 125, 190, 305, 495, and 760 m. Unlike the baroclinic transfer rate, the barotropic transfer rates from the surface down to the 125-m depth are about the same magnitude, and begin to decrease with depth below 125 m. The highest transfer rate occurs near (31.8°N , 78°W), immediately north of the bump. The magnitude of the barotropic transfer rate is $3 \times 10^{-7} \text{ m}^2 \text{ s}^{-3}$, which is also in agreement with the Dewar and Bane (1985) calculation based on observational data ($0.12 \times 10^{-2} \text{ ergs cm}^{-3} \text{ s}^{-1}$).

Since the barotropic and baroclinic energy transfer processes occur mainly above 125-m depth, in the following we will analyze the energy balance at 25-m and 125-m depth. The terms (1a) and (1b) are calculated at 25- and 125-m depth, and the results are shown in Figs. 10 and 11. The results indicate that barotropic transfer is the major contributor to the energy change at the 25-m depth, whereas baroclinic transfer is the major contributor to the energy change at 125-m depth. It is worth noting that, in Fig. 11a, the region south of the

CB is a weak energy loss region ($-1.0 \times 10^{-7} \text{ m}^2 \text{ s}^{-3}$), while the region immediately north of the bump is a region of strong energy growth ($8.0 \times 10^{-7} \text{ m}^2 \text{ s}^{-3}$), and the region farther north is a region of strong energy decay ($-5.0 \times 10^{-7} \text{ m}^2 \text{ s}^{-3}$). Lee et al. (1991), after analyzing the cross-front momentum and heat fluxes based on historical observation data, concluded that two stable regions in the SAB are from 30° to 32°N and from 33° to Cape Hatteras, and eddies grow rapidly after the Gulf Stream impinges upon the CB and turns offshore. The modeling results presented above are in agreement with the conclusion of Lee et al. (1991). The model results also show the presence of large differences between the cyclonic and anticyclonic flanks of the Gulf Stream. According to Dewar and Bane (1985), for most of the Gulf Stream, eddies on the cyclonic side of the flow perform net work on the mean flow, while those on the anticyclonic side are energized by the mean flow. This feature is consistent with the model results as shown in Figs. 10 and 11.

b. Analysis with MS-EVA for case IV

The multiscale energy and vorticity analysis is a new methodology for the investigation of multiscale interactive oceanic processes that are intermittent in space and time (Liang and Robinson 2005a; LR05). Through exploring pattern generation and energy transfers, transports, and conversions, the intricate relationships between events on varying scales and locations in phase space and physical space can be sorted out. The basic idea for this has been delineated and the formulation developed, by Liang and Robinson (2005a), and an avenue to application is established in LR05. The latter study also established a generalization of the concept of stability on a localized basis, which allows one to build an easy-to-use criterion for the identification of baroclinic and barotropic instability processes utilizing actual or modeled ocean and atmosphere datasets. Since our hypothesis is that the combined effect of the CB and the isobathic curvature in the region leads to the generation and amplification of the "Charleston Trough" through baroclinic and barotropic instability processes, we will use MS-EVA to analyze the dataset obtained from case IV.

For theory and detailed derivation of MS-EVA, please refer to Liang and Robinson (2005a) and LR05. A brief description of MS-EVA in a symbolic form and the definition of all terms in the equations are included in the appendix. The difference between MS-EVA and the traditional energy analysis method is that all MS-EVA terms are local in time and space, and hence the criterion is applicable to problems on a generic basis. The traditional energy analysis method uses an energy

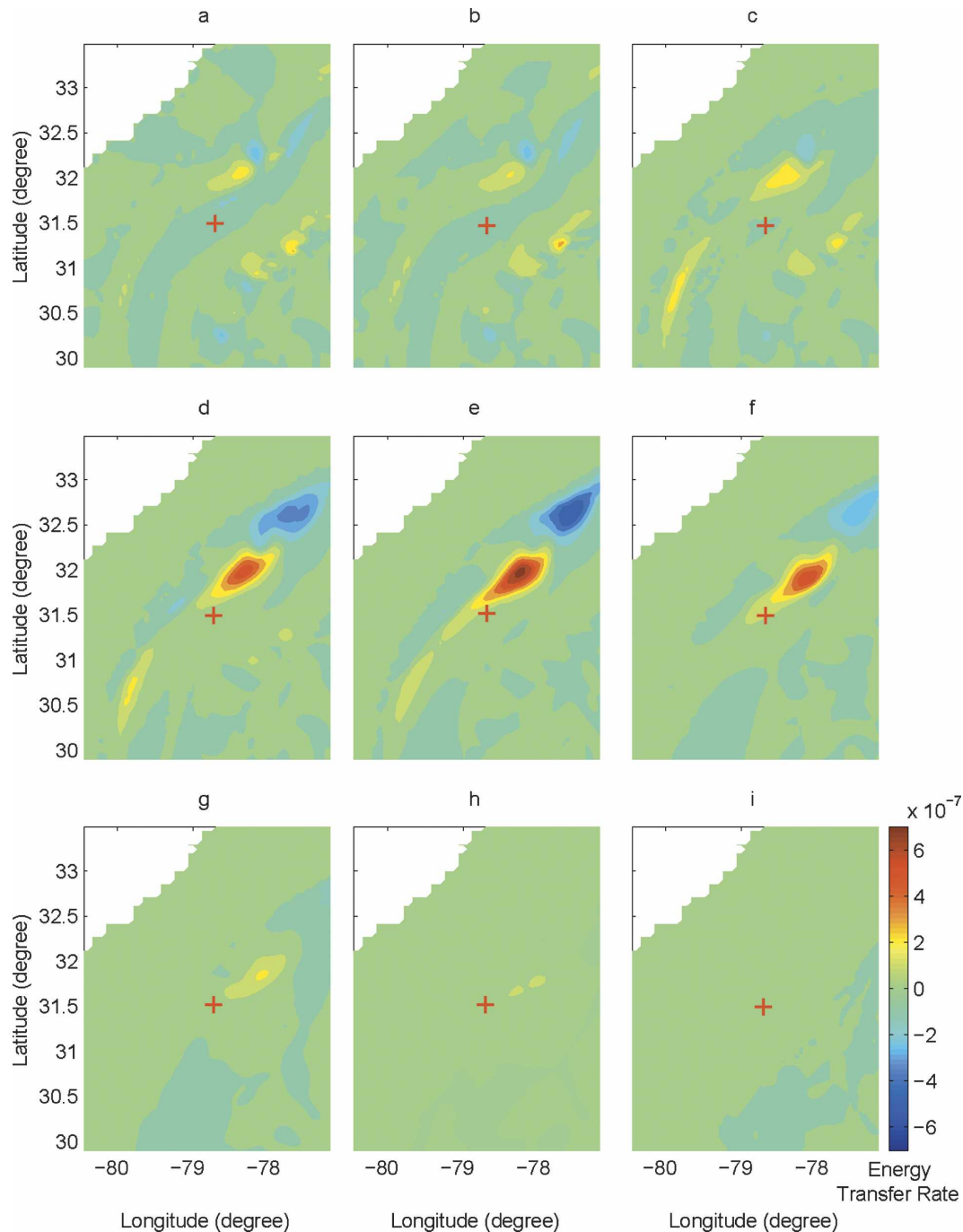


FIG. 8. Baroclinic transfer rate ($\text{m}^2 \text{s}^{-3}$) of case IV at depths 7.5, 25, 47.5, 80, 125, 190, 305, 495, and 760 m.

equation of either a time or a space average. However, with MS-EVA, we can compute energy balance for specific regions and time period.

The first step is to determine the scale windows to start the MS-EVA process. From moored or Eulerian current meter observations, Pietrafesa and Janowitz (1980) determined the periods of meanders and eddies

in the SAB to be 2–14 days. As we have previously found, a perturbation in the flow field occurred from 23 January to 3 February in case IV. For the MS-EVA process, we extracted a time series of 256 time steps with a time interval $\Delta t = 7200 \text{ s}$; therefore, the total length of study is 21.33 days from 12 January to 2 February. We choose $j_0 = 0$, $j_1 = 4$, and $j_2 = 8$ for the scale

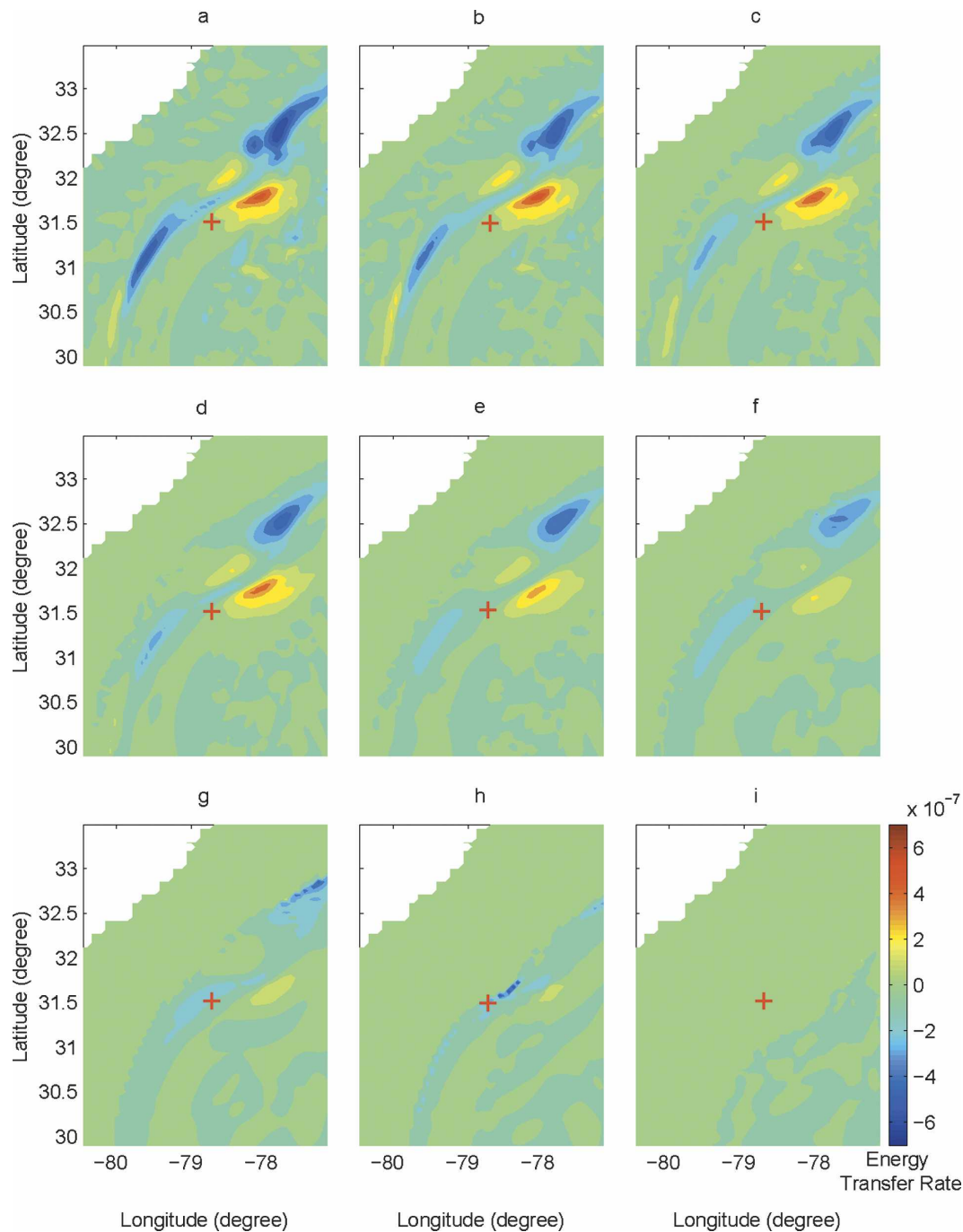


FIG. 9. As in Fig. 8 but for barotropic transfer rate of simulation IV.

window limitations. With this setting: the large-scale window covers variations with a time scale greater than 21.33 days ($j_0 = 0$); the mesoscale window covers variations from 1.33 days ($j_1 = 4$) to 21.33 days ($j_0 = 0$); and the submesoscale window covers variations from 0.083 days ($j_2 = 8$) to 1.33 days ($j_1 = 4$).

The MS-EVA setup also includes a preparation of

the background density profile and projection of data onto the model grid. We first interpolate linearly all field variables defined on density coordinates onto nine z levels: 7.5, 25, 47.5, 80, 125, 190, 305, 495, and 780 m. The stationary background density profile $\rho(z)$ is then calculated by averaging all the interpolated density data, and its vertical gradient ($\partial\rho/$

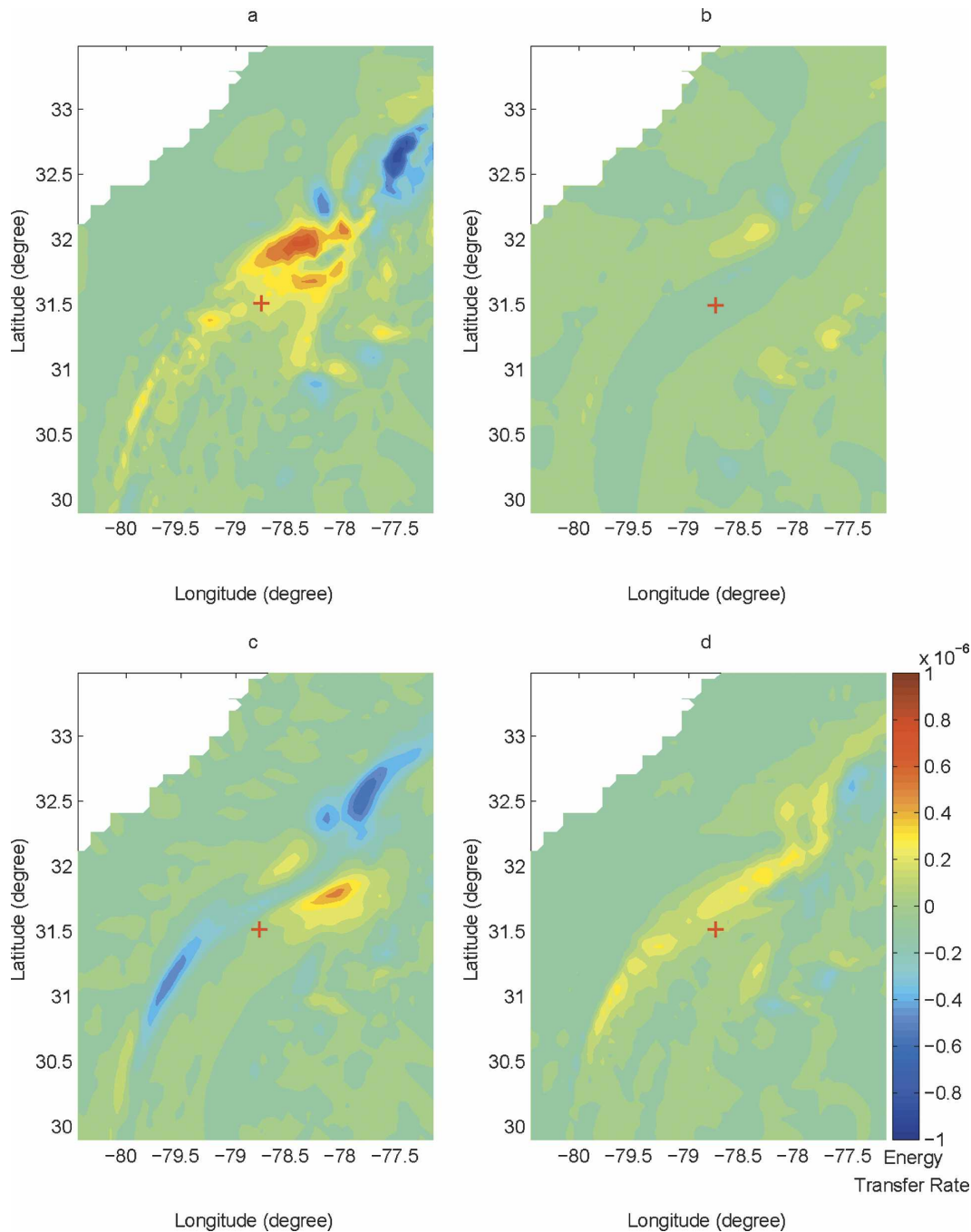


FIG. 10. Eddy energy budget of case IV at 25-m depth: (a) total rate change of energy, (b) baroclinic transfer rate, (c) barotropic transfer rate, and (d) pressure work (units: $\text{m}^2 \text{ s}^{-3}$).

∂z) is used to compute the Brunt–Väisälä frequency, which is needed for computing available potential energy.

Figures 12a1–a6 show the MS-EVA baroclinic transfer rate of case IV on 27 January at depths 25, 47.5, 80, 125, 190, 305, 495 m. We can see that the highest baroclinic transfer rate occurs at 125-m depth, and maxi-

mum transfer rate reaches $10^{-6} \text{ m}^2 \text{ s}^{-3}$ at (32°N, 78°W). The baroclinic transfer rate is small on the surface levels and bottom levels. Both the value and the location of the baroclinic transfer are in agreement with the traditional analysis shown in section 3a. Figures 12b1–b6 show the corresponding barotropic transfer rate, and it is large from the surface level to 125-m depth and

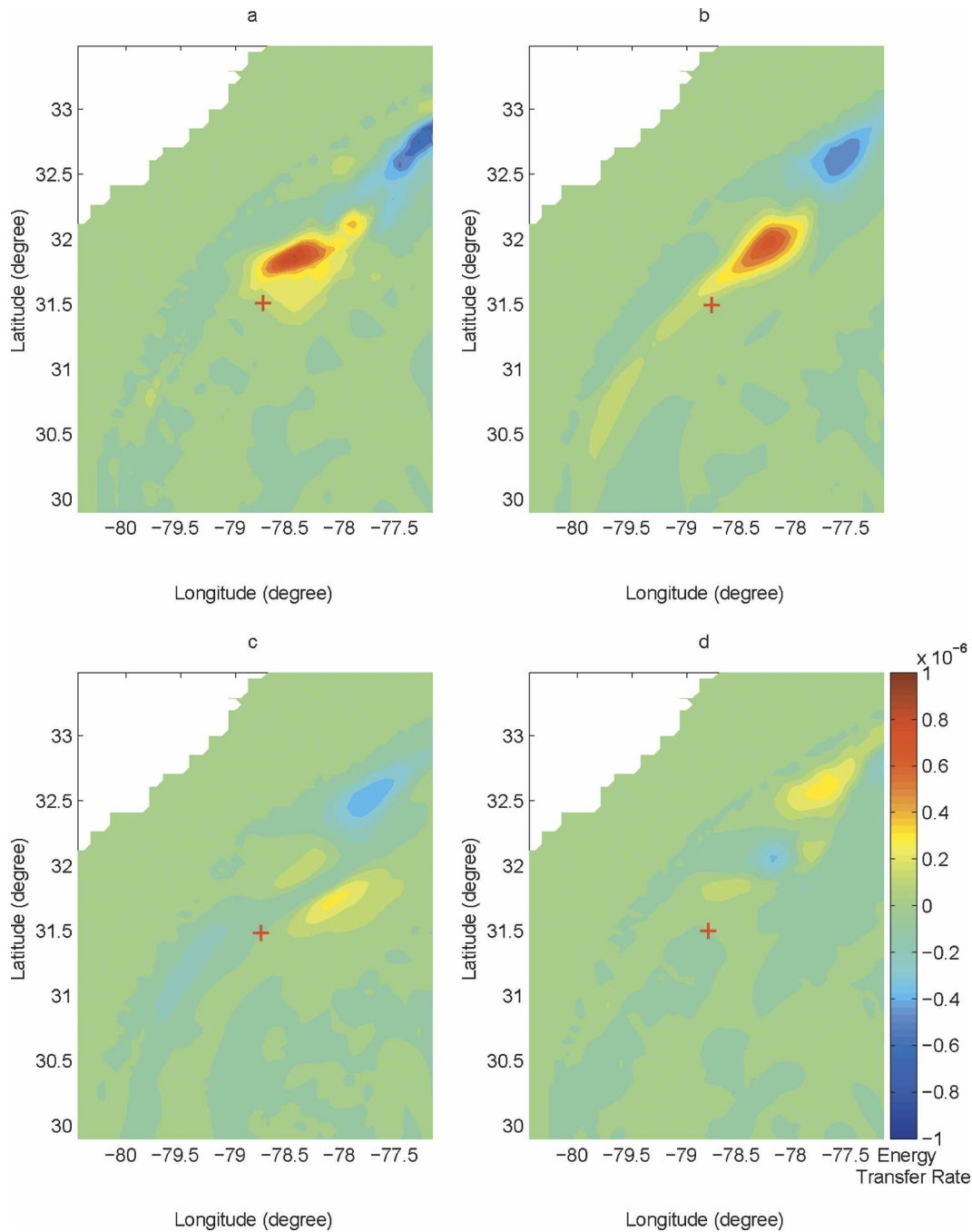


FIG. 11. Eddy energy budget ($\text{m}^2 \text{s}^{-3}$) of case IV at 125-m depth: (a) total rate change of energy, (b) baroclinic transfer rate, (c) barotropic transfer rate, and (d) pressure work.

small below 125 m. This is also in agreement with the traditional analysis described in the previous section (4a).

Since both baroclinic and barotropic transfer rates are large at the 125-m depth, we will look at the interaction between the perturbation and mean flow at this depth for the meander event that occurred between 23 and 31 January. Figure 13 shows the evolution of the

baroclinic transfer rate from 23 to 21 January. It shows that the baroclinic transfer rate north of the CB increased from 23 to 26 and decreased from 27 to 31 January. Figure 14 shows the evolution of the barotropic transfer rate from 23 to 31 January. During the event, the barotropic transfer rate downstream of the CB near 32°N was positive, and large values occurred

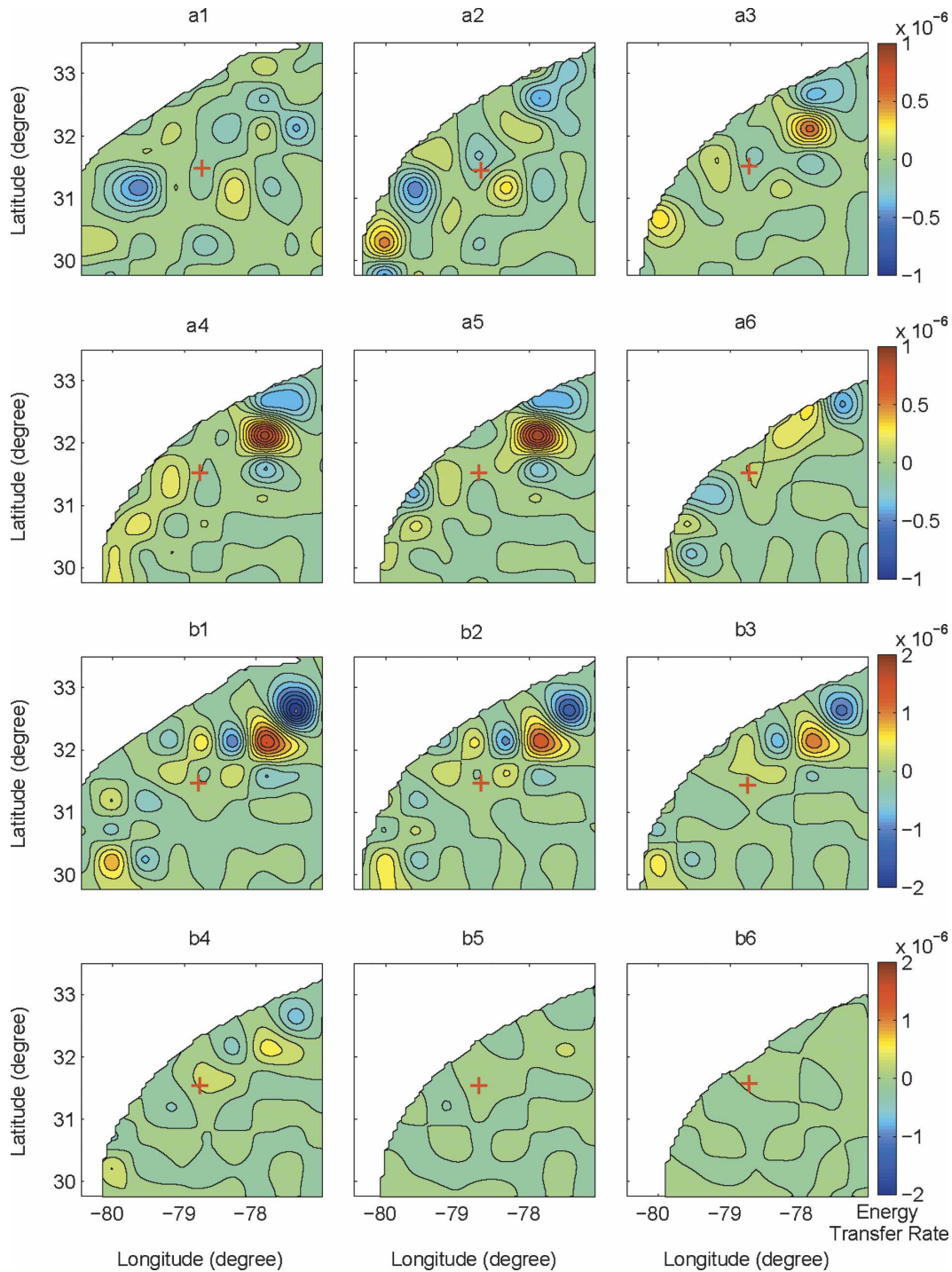


FIG. 12. MS-EVA baroclinic transfer rate of case IV on 27 Jan at (a1) 25-, (a2) 47.5-, (a3) 80-, (a4) 125-, (a5) 190-, and (a6) 305-m depth and barotropic transfer rate of case IV on 27 Jan at (b1) 25-, (b2) 47.5-, (b3) 80-, (b4) 125-, (b5) 190-, and (b6) 305-m depth (units: $\text{m}^2 \text{s}^{-3}$).

on 26 and 29 January. These indicate that mixed instability is involved in this eddy amplification process.

Regarding the local energy balance, Figs. 15a and 15b depict the time series of eddy available potential energy

(APE) and its budget at (32.2°N, 77.9°W), respectively. Large eddy APE occurred on 25 and 29 January, and a relative minimum occurred on 27 January (Fig. 15a). This is mainly because of the northward propagation of

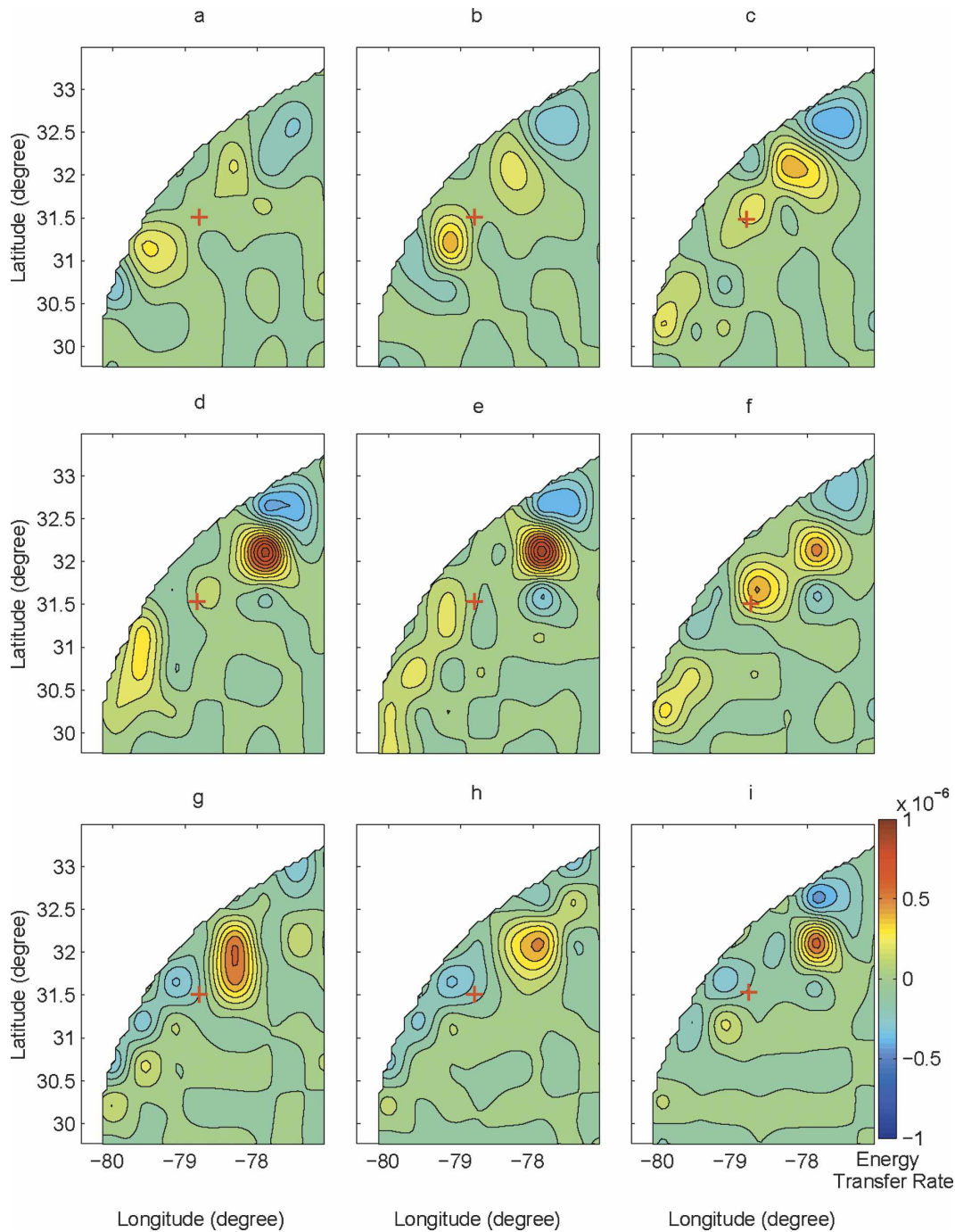


FIG. 13. Baroclinic transfer rate of case IV at 125-m depth from 23 to 31 Jan (units: $\text{m}^2 \text{s}^{-3}$); (a)–(i) correspond to 23–31 Jan, respectively.

the cold and warm anomalies and the associated energy transfer. As shown in Fig. 15b, the local change of APE is mainly due to alongstream advection processes associated with the northward propagation of the temperature anomalies. The baroclinic transfer rate (solid line) is the largest, but it is largely balanced buoyancy con-

version (dash line). As a result, the local change of APE (dash-dot line) is in close agreement with the advection of APE (dot line). The fact that the baroclinic transfer is mainly balanced by buoyancy conversion suggests that most of the APE transferred from mean flow to the eddy by the baroclinic process is converted into eddy

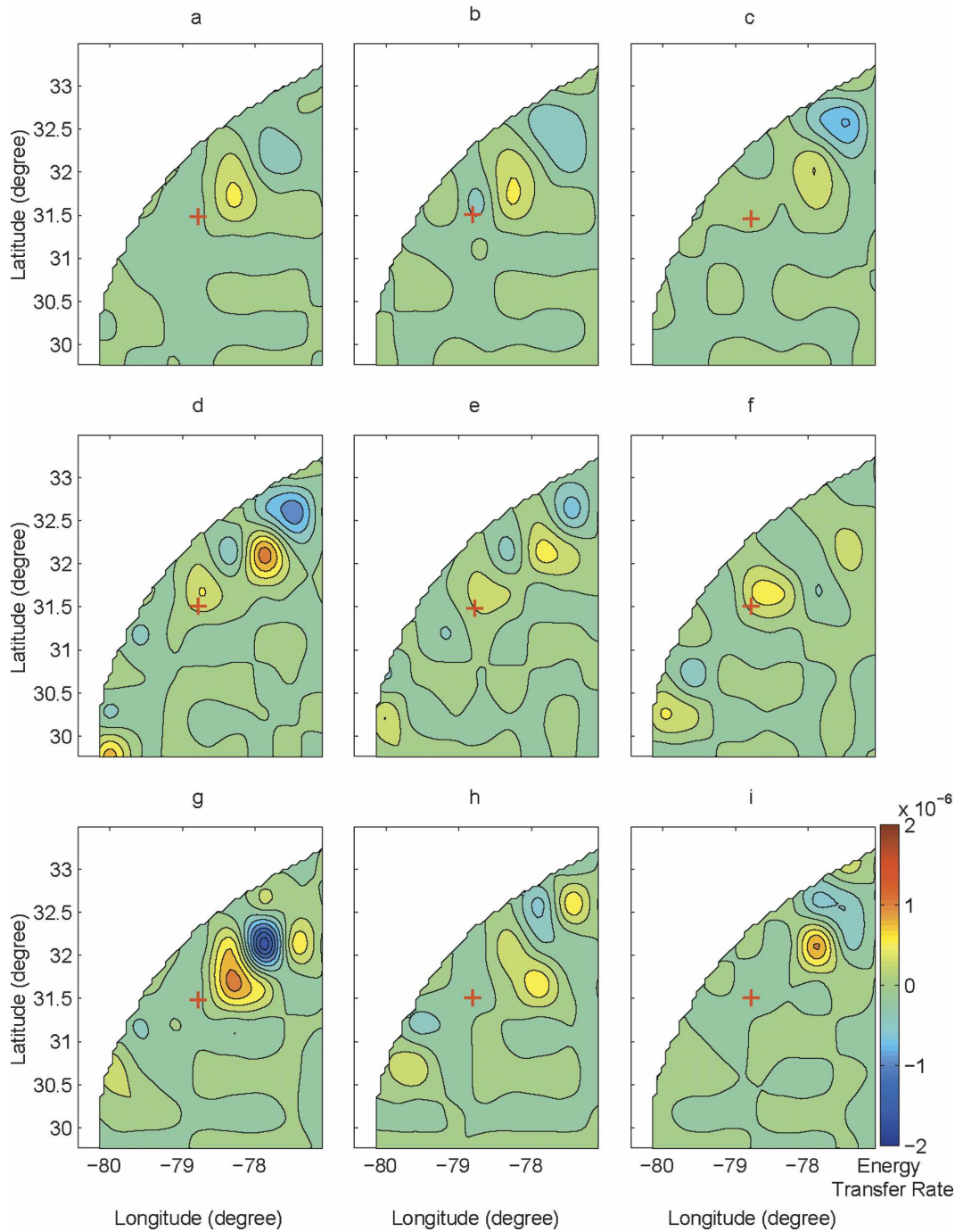


FIG. 14. As in Fig. 13 but for barotropic transfer rate.

kinetic energy through the buoyancy conversion process. When comparing Fig. 15a with Fig. 15b, it should be noted that the baroclinic process transfers energy into eddy APE, but the local eddy APE is decreasing because of advection between 25 and 27 January. This is further illustrated by Fig. 16. Figure 16a shows the times series of the eddy kinetic energy (KE) at the same

location as in Fig. 15. It shows that the eddy KE increased significantly from 19 to 29 January. A relative peak occurred on 25 January, and a minimum on 27 January, coinciding with the corresponding peak and minimum of eddy APE (Fig. 15a), suggesting that mean flow supplied both APE and KE into the eddy. Some of the eddy KE was transferred from the eddy APE. On

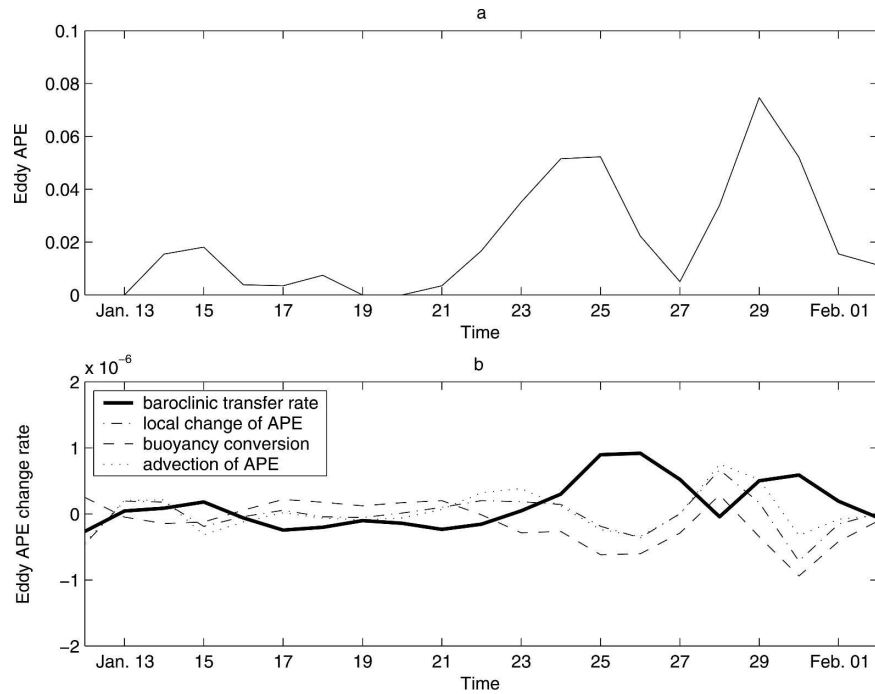


FIG. 15. (a) Eddy available potential energy (APE, units: $\text{m}^2 \text{s}^{-2}$); (b) APE budget (units: $\text{m}^2 \text{s}^{-3}$) at $(32.2^\circ\text{N}, 77.9^\circ\text{W})$.

closer examination of the KE transfer processes Fig. 16b), one can find that both the barotropic and buoyancy processes increased the KE, whereas the advection and pressure work processes led to a decrease in

the KE. The sum of these terms (local change of KE, dash-dot line) was mostly positive from 19 to 28 January so that the total energy at the study location increased with time during this period.

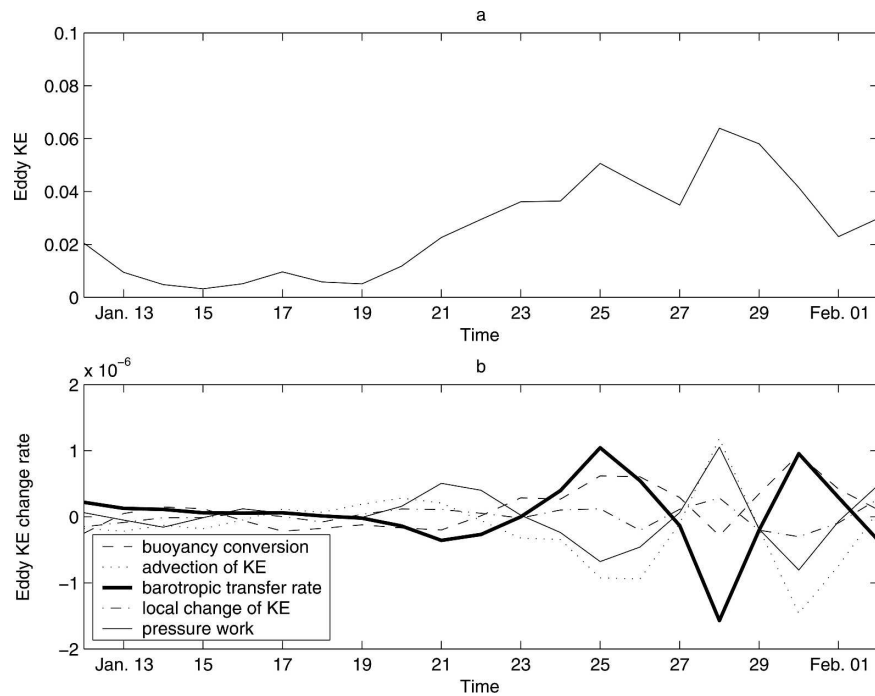


FIG. 16. As in Fig. 15 but for eddy kinetic energy (KE).

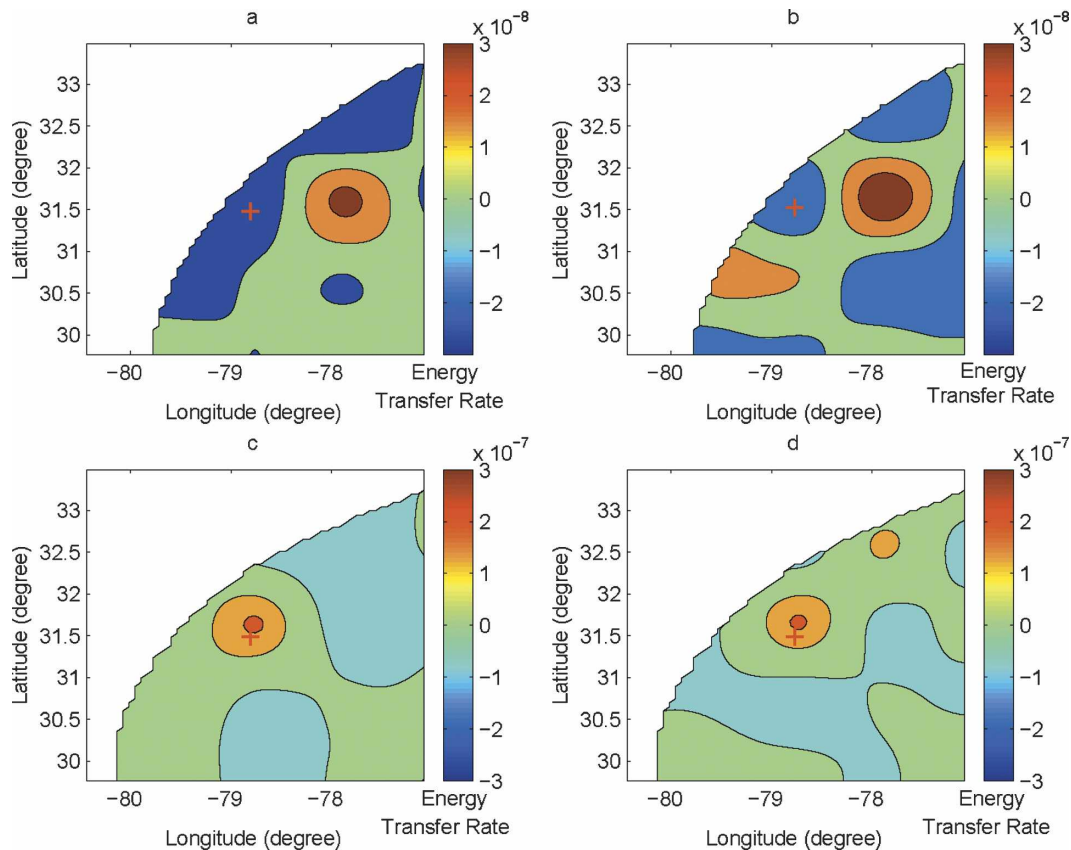


FIG. 17. Baroclinic and barotropic transfer rates at the 125-m depth on 25 Jan of cases II and III: (a) barotropic transfer rate of case II, (b) baroclinic transfer rate of case II, (c) barotropic transfer rate of case III, and (d) baroclinic transfer rate of case III (units: $\text{m}^2 \text{s}^{-2}$).

c. MS-EVA analysis for cases II and III

As discussed in the last section, amplification of meanders and perturbations in case IV are due to mixed baroclinic and barotropic energy transfers that occurred in regions with large isobathic curvature and downstream of the CB. To further quantify the contribution from the CB and the isobathic curvature, in this section, we present the results of MS-EVA analysis in cases II and III, and the results will be compared with that of case IV. MS-EVA is applied to cases II and III with the same settings as in case IV. Since barotropic and baroclinic transfer terms, Eqs. (4) and (5), are the indicators of the Gulf Stream instability, we will mainly compare the results of these two terms with those of case IV. Because the stratification is the same as in case IV, we will only look at the barotropic and baroclinic transfer rates at 125-m depth.

The baroclinic and barotropic transfer rates at the 125-m depth on 25 January of cases II and III are shown in Fig. 17. In case II, there are both baroclinic and barotropic energy transfers from the mean flow to the

eddy (Figs. 17c and 17d). However, the magnitude of the transfer rate is $3 \times 10^{-8} \text{ m}^2 \text{s}^{-3}$, much smaller compared with that in case IV. In case II, although there is a bump in the bathymetry, isobathic curvature is small compared with that in case IV. Therefore, the curvature of the isobaths seems to play a very important role in both baroclinic and barotropic processes. In case III, the bump is removed from the bathymetry, but the isobathic curvature is the same as in case IV. From Figs. 17a and 17b, we can see that there are barotropic and baroclinic energy transfers from the mean flow to the eddy in case III, and the magnitude of the energy transfer rate is about $3 \times 10^{-7} \text{ m}^2 \text{s}^{-3}$ (Figs. 17a and 17b). Although this energy transfer rate is smaller than that in case IV, it is much larger than for case II, which indicates that isobathic curvature alone can contribute more to the baroclinic and barotropic energy transfer processes than does the bump effect.

However, the amplification of the Gulf Stream meander/eddy in case IV is due to the combined effect of isobathic curvature and the bump. Neither isobathic curvature nor the bump alone will cause large enough

amplification to produce the Charleston trough, as seen in case IV. Intuitively, a frontal perturbation behaves differently when passing over a bump on a large curve than from passing over a bump on a small curve. The stability analyses presented in previous sections consistently indicate that the combined effects of isobathic curvature and the CB on baroclinic and barotropic instability processes are what are needed to explain the meander/eddy growth previously reported in the literature.

5. Discussions and conclusions

The effect of the isobathic curvature on the development and evolution of Gulf Stream frontal waves (meanders and eddies) in the vicinity of the CB is studied using HYCOM. Baroclinic and barotropic energy transfers from the Gulf Stream to its meanders and eddies are computed using the traditional energy balance equation as well as the MS-EVA method for four different cases. In case I, the curvature of the isobaths is artificially reduced and the CB is removed from the bathymetry. In this simulation, the simulated Gulf Stream meanders are barely noticeable in the study region. Energy transfer from the Gulf Stream to meanders and eddies is negligible. In case II, the curvature of the isobaths is kept the same as in case I, but a topographic rise of the scale of the CB is added to the bathymetry. In this simulation, Gulf Stream meanders that appear as cold and warm anomalies are amplified after passing over the bump. The total baroclinic and barotropic energy transfer rate in the CB region is on the order of $3 \times 10^{-8} \text{ m}^2 \text{ s}^{-3}$. In case III, the bathymetry contained realistic isobathic curvature similar to that of the actual bathymetry in the region, which is larger than that of cases I and II. However, the CB is removed from the bathymetry as in case I. In this simulation, large meanders are simulated, but the development of these meanders was not confined to the region of the downstream region of the CB. The total baroclinic and barotropic energy transfer rate in this case (approximately $3 \times 10^{-7} \text{ m}^2 \text{ s}^{-3}$) is an order of magnitude greater than in case II, suggesting that isobathic curvature can generate Gulf Stream meanders and eddies even without the presence of the CB. In case IV, actual bathymetry data that contains both the “CB” and the isobathic curvature was used. In this case, large-amplitude Gulf Stream meanders were simulated and there was also a tendency for the amplification of the meanders to be anchored downstream of the CB. For case IV, the simulated amplitude as well as the location of the simulated Gulf Stream meander in the CB region is consistent with observations.

Case III, which has a larger isobathic curvature but

no bump, produced larger amplitudes in the Gulf Stream meanders than those seen in cases I and II. A question that needs to be answered is whether the smaller amplitude of the Gulf Stream meanders seen in cases I and II as compared with that in case III is indeed due to the smaller isobathic curvature in cases I and II or caused by the difference of the stabilization effect associated with the isobathic slope among the cases. Might the larger isobathic curvature in case III keep the Gulf Stream in deeper water and thus farther away from the large continental slope than in cases I and II? If so, the Gulf Stream in case III may be more unstable than in cases I and II simply due to the smaller bottom topographic slope. A careful diagnosis of the model results indicates that this is not the case. The Gulf Stream position is self-adjusted to the bathymetry in all cases. In cases I and III, the Gulf Stream axis followed the 550-m isobath closely. As a result, the stabilization effect of the isobath slope remains more or less the same in both cases I and III.

To further quantify the stabilization effect in cases I and III, the potential vorticity balance is analyzed using the method suggested by Pickart and Watts (1993). Figures 18a and 18b shows the bottom layer potential vorticity balance among local time rate of change, along-stream advection, and cross-stream advection in cases I and III at the 550-m isobath at 31°N. In both cases, the local time rate of change is mainly balanced by the along-stream advection term. This is different from the case studied by Pickart and Watts (1993) where the local time rate of change term is mainly balanced by the cross-stream advection term since the Gulf Stream meanders were over steep topography in their case. In both cases I and III, the cross-stream advection terms are small compared with the other terms in the balance. This is because the slope of the isobaths and isopycnic slope are inclined at the same angle so that cross-stream movement of the bottom layer does not cause much squeezing or stretching of the water column. The magnitude of the cross-stream advection term in cases I and III are about the same ($0.2 \times 10^{-11} \text{ s}^{-2} \text{ m}^{-1}$), which indicates that the stabilization effect of the slope of the isobaths are about the same in the two cases. Therefore, the larger meander in case III is not the result of the lateral drift of the Gulf Stream, but rather, the isobathic curvature.

The results from this study suggest that the formation of the Charleston Trough, a Gulf Stream meander that appears as a low pressure or depressed water surface region downstream of the bump, is the result of the combined effect of the CB and the isobathic curvature in the region. The isobathic curvature plays a major role in enhancing the baroclinic and barotropic energy

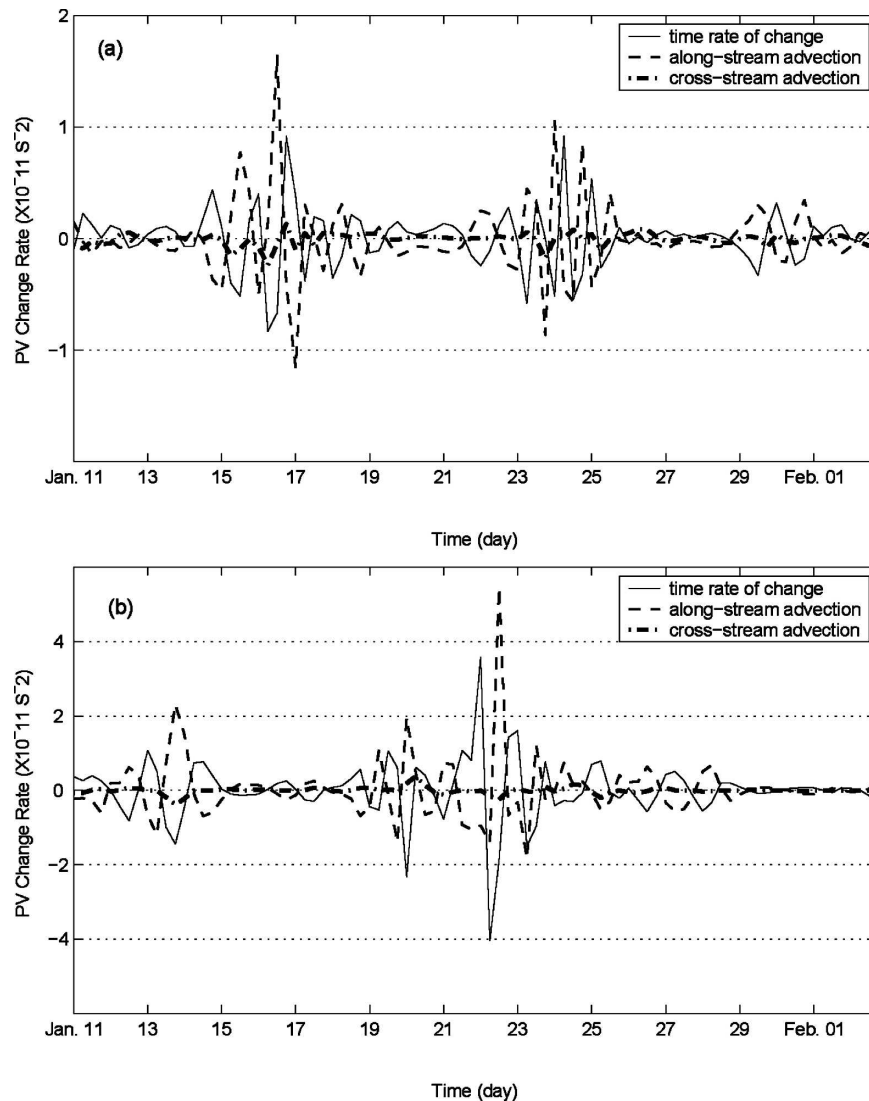


FIG. 18. Potential vorticity balance: (a) local time change (solid line), along-stream advection (dashed line), and cross-stream advection (dash-dot line) for case I; (b) as in (a) but for case III (units: s^{-2}).

transfer rates, whereas the bump provided a localized mechanism to maximize the energy transfer rate over the CB. In our simulations, the maximum baroclinic transfer rate occurred at the 125-m depth. However, the barotropic transfer rate was large and uniform above 125 m but was relatively small below.

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stability diagnostic program MS-EVA and Alan Wallcraft for assisting with the configuration of HYCOM at NCSU. The constructive suggestions from the two anonymous reviewers are also highly appreciated.

APPENDIX

A Brief Description of MS-EVA Method

The MS-EVA is a methodology for the investigation of multiscale interactive oceanic processes that are intermittent in space and time. For detail theory and derivation of MS-EVA, please refer to Liang and Robinson (2005a) and LR05. A brief description of MS-EVA in a symbolic form and all terms in the equations will be

included in this appendix. In the MS-EVA, processes are represented on scale windows. A scale window implies a subspace of the space to which the field under consideration belongs with a certain range of the scales involved. A scale level j is a dimensionless index such that 2^{-j} measures the passage of the events since the beginning for a time series scaled by its duration. We need three scale levels, j_0 , j_1 , and j_2 , $j_0 \leq j_1 \leq j_2$, that demarcate three mutually exclusive windows: 1) large-scale ($j \leq j_0$); 2) mesoscale ($j_0 < j \leq j_1$); and 3) submesoscale ($j_1 < j \leq j_2$). For simplicity, a window may be referenced as $\hat{w}$, with $\hat{w} = 0, 1, 2$ standing for largescale, mesoscale, and submesoscale, respectively. MS-EVA provides a way to study the interactions between these windows.

In a symbolic form, the growths of kinetic energy $K_n^{\hat{w}}$ and available potential energy $A_n^{\hat{w}}$ on window $\hat{w}$ ($\hat{w} = 0, 1, 2$) and at time step n for a frictionless fluid flow are governed by

$$K_n^{\hat{w}} = \Delta Q_{K_n^{\hat{w}}} + \Delta Q_{P_n^{\hat{w}}} + T_{K_n^{\hat{w}}} - b_n^{\hat{w}} \quad \text{and} \quad (\text{A1})$$

$$A_n^{\hat{w}} = \Delta Q_{A_n^{\hat{w}}} + T_{A_n^{\hat{w}}} + b_n^{\hat{w}}, \quad (\text{A2})$$

where

$$T_{K_n^{\hat{w}}} = T_{K_n^{\hat{w}},h} + T_{K_n^{\hat{w}},z} \quad \text{and} \quad (\text{A3})$$

$$T_{A_n^{\hat{w}}} = T_{A_n^{\hat{w}},h} + T_{A_n^{\hat{w}},z} + TS_{A_n^{\hat{w}}}. \quad (\text{A4})$$

In the equations, the ΔQ terms represent the multiscale transport process on the specified scale window, “ b terms” represent buoyancy conversion terms, and the “ T terms” are perfect transfers among scale windows in the sense that they vanish when averaged over windows $\hat{w}$ and time step n . The definition of the symbols for multiscale energetics at scale window $\hat{w}$ and time step n in Eqs. (2)–(5) are listed below:

$K_n^{\hat{w}}$	Time rate of change of KE
$\Delta Q_{K_n^{\hat{w}}}$	KE advective working rate
$T_{K_n^{\hat{w}},h}$	KE transfer due to horizontal flow
$T_{K_n^{\hat{w}},z}$	KE transfer due to vertical flow
$-b_n^{\hat{w}}$	Rate of buoyancy conversion
$\Delta Q_{P_n^{\hat{w}}}$	Pressure working term
$A_n^{\hat{w}}$	Time rate of change of APE
$\Delta Q_{A_n^{\hat{w}}}$	APE advective working rate
$T_{A_n^{\hat{w}},h}$	APE transfer due to horizontal gradient of ρ
$T_{A_n^{\hat{w}},z}$	APE transfer due to vertical gradient of ρ
$b_n^{\hat{w}}$	Rate of inverse buoyancy conversion
$TS_{A_n^{\hat{w}}}$	APE transfer due to $\partial\bar{p}/\partial z$

Perfect transfer is a key concept in the MS-EVA formulation (Liang and Robinson 2005a). It allows one to separate transport processes from the nonlinear energetic terms based on a firm physical ground and, hence, to tell whether the energy growth for a window at a

particular location and time is due to the local energy transfer or transport from surrounding regions.

A natural generalization of these stability theories to handle real-world problems is fulfilled with these terms. In the simple case with only two windows (window 0 and window 1), let

$$\begin{aligned} \text{BC} &= T_{A_n^1}^{0 \rightarrow 1} = T_{A_n^1,h}^{0 \rightarrow 1} + T_{A_n^1,z}^{0 \rightarrow 1} + TS_{A_n^1}^{0 \rightarrow 1} \quad \text{and} \\ \text{BT} &= T_{K_n^1}^{0 \rightarrow 1} = T_{K_n^1,h}^{0 \rightarrow 1} + T_{K_n^1,z}^{0 \rightarrow 1}, \end{aligned} \quad (\text{A5})$$

where the superscript $0 \rightarrow 1$ denotes the energy transfer from window 0 (large scale) to window 1 (mesoscale), and T_A terms represent potential energy transfer and T_K terms represent kinetic energy transfer. The subscripts “ h ” and “ z ” denote the energy transfer in the vertical and horizontal direction respectively. A criterion was derived in Liang and Robinson (2005a) for instability identification:

- 1) a flow system is locally unstable if $\text{BT} + \text{BC} > 0$, and vice versa;
- 2) for an unstable system, if $\text{BT} > 0$ and $\text{BC} < 0$, the instability the system undergoes is barotropic;
- 3) for an unstable system, if BC is positive but BT is not, then the instability is baroclinic; and
- 4) if both BT and BC are positive, the system must be undergoing a mixed instability.

For convenience, BC and BT may be referred to as, respectively, the baroclinic instability indicator and the barotropic instability indicator, though neither of them per se is enough for instability identification.

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